Question 1: USART (8251A) Communication Design

The **USART (8251A)** acts as a mediator between the parallel bus of the microprocessor and serial I/O devices.

Practical Differences in Design:

1. Synchronous Mode:
    - **Clocking:** Requires a common clock signal between the transmitter and receiver.
    - **Framing:** Uses "Sync Characters" instead of start/stop bits. The system must be designed to detect these characters to establish a block boundary.
    - **Efficiency:** Higher data rate because it doesn't use overhead bits for every byte.
2. Asynchronous Mode:
    - **Clocking:** The transmitter and receiver have independent clocks (usually 16x or 64x the baud rate).

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I/O devices.

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    - **Efficiency:** Higher data rate because it doesn't use overhead bits for every byte.
2. Asynchronous Mode:
    - **Clocking:** The transmitter and receiver have independent clocks (usually 16x or 64x the baud rate).
    - **Framing:** Each byte is "framed" by a **Start bit** (low) and one or two **Stop bits** (high).
    - **Practical Use:** Ideal for low-speed communication where the sender and receiver are not synchronized.

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Question 2: I/O Interfacing & 8255A PPI

(a) I/O Interface Block Diagram

An interface consists of:

- **Address Decoder:** Selects the specific I/O port when the CPU places an address on the bus.
- **Data Buffer/Latches:** Temporarily holds data being transferred to/from the CPU.
- **Control Logic:** Manages RD and WR signals to ensure data is read or written at the correct time.

(b) 8255A Modes of Operation

1. **Mode 0 (Basic I/O):** All ports (A, B, and C) function as simple input or output ports without "handshaking."
2. **Mode 1 (Strobed I/O):** Ports A and B use Port C pins as control signals (handshaking) to manage data transfer.

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3. **Mode 2 (Bidirectional Bus):** Only Port A can be used in this mode. It allows two-way communication on a single 8-bit bus with handshaking signals from Port C.

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Question 3: Instruction Sets & Assembly

(a) Groups of Instructions

- **Data Transfer:** MOV, MVI, LXI, LDA, STA.
- **Arithmetic:** ADD, SUB, INR, DCR.
- **Logical:** ANA, ORA, XRA, CMP, RAR.
- **Branching:** JMP, JC, JNZ, CALL, RET.
- **Machine Control:** HLT, NOP, EI, DI.

(b) 16-bit Addition Program (8085)

asm

```
LHLD 2000H    ; Load first 16-bit
number from 2000H & 2001H into HL
XCHG          ; Move it to DE
LHLD 2002H    ; Load second 16-bit
number from 2002H & 2003H into HL
DAD D         ; Add DE to HL (Result in
HL)
SHLD 2004H    ; Store the 16-bit result
at 2004H & 2005H
HLT           ; Terminate
```

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```
SHLD 2004H     ; Store the 16-bit result
at 2004H & 2005H
HLT            ; Terminate
```

(c) Program Function Trace

The provided code performs **Bit Counting (counting 1s)**:

1. **MVI B, 00H**: Initializes counter for 1s.
2. **MVI C, 08H**: Loop counter (8 bits in a byte).
3. **MOV A, D**: Copies the number to be checked from register D to Accumulator.
4. **BACK: RAR**: Rotates the Accumulator right through the Carry flag.
5. **JNC SKIP**: If the bit shifted into Carry is '0', it skips the increment.
6. **INR B**: If the bit was '1', increment register B.
7. **DCR C / JNZ BACK**: Repeats the process 8 times.
   Final Result: Register B contains the total count of '1' bits present in the original byte in register D.

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Question 4: T-states & Timing Diagrams

Based on the provided diagram (Opcode Fetch & Memory Read):

- **T1 State:** The microprocessor places the 16-bit address on the bus. **ALE** (Address Latch Enable) goes high to latch the lower address bits.
- **T2 State:** The CPU activates the $\overline{RD}$ (Read) signal. The memory/IO device prepares the data. The CPU checks the **READY** pin for wait states.
- **T3 State:** Data is placed on the data bus and is read into the CPU's internal register. The $\overline{RD}$ signal goes high (inactive) at the end.
- **T4 State:** Internal to the CPU. The opcode is decoded and the instruction is executed.

Question 5: 8259A Programmable Interrupt Controller (PIC)

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The **8259A** is essential for expanding the interrupt capabilities of a microprocessor (like the 8085 or 8086).

Justification of Use:

1. **Expansion:** A standard 8085 has limited hardware interrupts. One 8259A handles 8 interrupts, and they can be cascaded to handle up to 64.
2. **Priority Resolution:** It contains a **Priority Resolver** that decides which interrupt to service first if multiple requests arrive simultaneously.
3. **Vectoring:** It provides the "Vector Address" to the CPU, telling it exactly where the Interrupt Service Routine (ISR) is located in memory.
4. **Modes:** It supports various modes like Fully Nested Mode, Automatic Rotation, and Interrupt Masking (via the Interrupt Mask Register - IMR).

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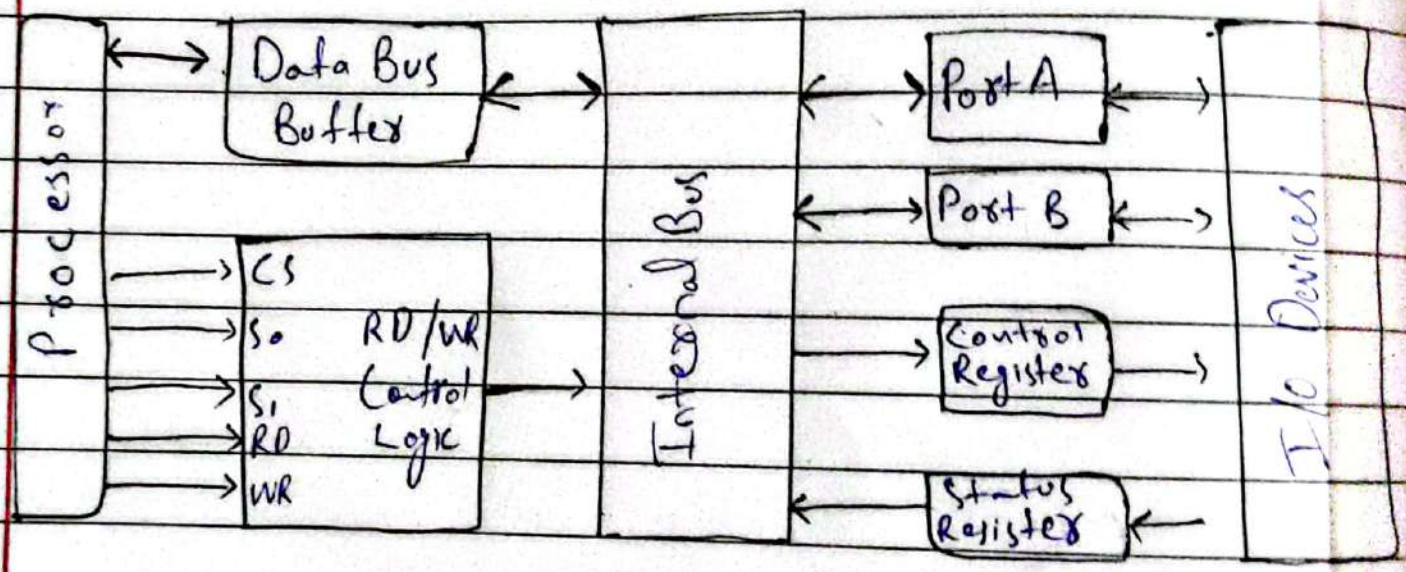
Date:

Past Paper:

Q2 a) Block Diagram: